

23CSE202 DATABASE MANAGEMENT SYSTEMS

S3 B.Tech CSE

Department of Computer Science and Engineering

Amrita School of Computing, Amritapuri Campus

PROJECT ABSTRACT

Group No : C13

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Project Title : Campus Canteen Pre-Order and Queue Management System

Abstract: Waiting in a long queue during lunch breaks is a common problem in most college canteens. Students often waste their valuable time waiting to place orders, while the canteen staff struggle to manage large crowds and sudden rush hours. Since ordering is usually done only at the counter, there is no way for students to preorder food, know when it will be ready, or estimate how long they will have to wait. This often leads to overcrowding, inefficient food preparation, and reduced student satisfaction.

The **Campus Canteen Pre-Order and Queue Management System** is a DBMS based application designed to solve these issues by allowing students to place food orders in advance and track their order status in real time. The system also estimates waiting time based on the current kitchen workload and helps the canteen staff manage orders more efficiently.

The main objective of this project is to design a relational database that stores and manages student details, menu items, food orders, order status, and kitchen capacity. It also automates wait-time estimation, maintains an organized order queue, and provides useful reports about order trends and peak hours.

The database will be implemented using **PostgreSQL**. It will contain well-normalized tables such as **Students**, **MenuItems**, **Orders**, **OrderItems**, **OrderStatusLog**, and **KitchenCapacity**. The database

Key Features

- Students can browse the menu and place food orders before reaching the canteen.
- The system calculates an estimated waiting time based on the number of pending orders, preparation time of each food item, and current kitchen capacity using a stored procedure.
- Whenever an order status changes (Placed → Preparing → Ready → Collected), a trigger automatically records the status change and timestamp in the **OrderStatusLog** table.
- Orders are prioritized based on their booking time or selected pickup slot so that preorders are processed before similar walkin orders.
- Menu items automatically become unavailable once their daily stock reaches zero.
- Different users have different access permissions. Students can only view their own orders, while canteen staff can monitor and update all active orders through a queue dashboard. The admin manages menu items, prices, and daily stock.
- Order placement is transaction safe. If an item becomes unavailable while placing an order, the entire transaction is rolled back to maintain data integrity.
- The system generates reports such as peak ordering hours, most popular food items, daily sales, and average preparation time using SQL joins and aggregate queries.

Users of the System

The system is designed for three types of users:

- **Students** :Place food orders, check order status, and view order history.
- **Canteen Staff**: Process incoming orders and update preparation status.
- **Admin**: Manage menu items, pricing, stock availability, and monitor overall system performance.